

ROEBOURNE AP, Australia

WMO: 943080

Lat: 20.759S

Lon: 117.158E

Elev: 37

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	49.2	51.8	21.3	16.0	73.2	24.6	18.7	72.8	25.5	71.8	23.6	72.6	4.3	210	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.4	108.5	71.5	106.0	71.8	103.8	72.4	81.0	92.8	80.3	91.6	79.6	90.8	14.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.5	85.6	77.7	144.1	85.3	76.9	140.1	85.2	44.9	93.2	44.0	91.7	43.3	90.7	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.0	22.7	20.7		DB	44.5	114.6	2.9	2.8	42.3	116.6	40.6	118.2	39.0	119.8	36.8	121.8
				WB	38.3	83.2	2.6	1.2	36.4	84.0	34.9	84.7	33.4	85.3	31.5	86.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	89.4	89.5	88.3	84.0	75.3	69.1	68.1	70.1	75.4	83.0	86.5	89.1	(6)
	DBStd	9.21	4.26	4.11	4.38	4.00	4.23	4.13	4.12	3.69	4.31	5.41	4.67	4.37	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	31	0	0	0	0	0	11	16	4	0	0	0	0	(9)
	CDD50	11169	1220	1106	1188	1019	785	572	562	624	762	1024	1096	1211	(10)
	CDD65	5727	755	686	723	569	321	133	114	163	312	559	646	746	(11)
	CDH74	76432	10835	9953	10404	7299	3203	1092	1191	1932	3703	7456	8782	10582	(12)
	CDH80	43721	6532	6068	6238	4056	1397	231	293	640	1839	4488	5380	6559	(13)
(14) Wind	WSAvg	10.6	12.4	11.4	10.5	8.9	9.1	9.6	9.2	9.5	10.3	11.4	12.4	12.7	(14)
(15) Precipitation	PrecAvg	11.7	2.3	3.1	2.0	0.6	1.0	1.4	0.6	0.2	0.0	0.0	0.0	0.4	(15)
	PrecMax	25.8	12.9	11.9	12.7	4.5	5.8	12.2	3.3	1.1	0.3	0.2	0.5	3.9	(16)
	PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.1	3.4	3.1	2.8	1.1	1.4	2.4	0.9	0.4	0.1	0.0	0.1	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.2	110.8	109.0	104.2	96.5	88.2	89.4	92.3	100.2	107.4	109.7	111.3	(19)
		MCWB	73.8	74.2	72.0	69.9	65.3	65.2	61.9	62.6	64.8	67.4	68.7	70.8	(20)
	2%	DB	105.8	106.6	105.4	101.1	93.2	85.2	86.0	88.6	96.3	104.4	106.4	107.6	(21)
		MCWB	75.8	75.0	72.2	69.5	65.3	62.5	61.3	62.1	63.8	67.4	69.3	70.9	(22)
	5%	DB	102.5	103.1	102.4	98.3	90.4	82.7	83.2	86.1	93.1	101.4	103.3	104.2	(23)
		MCWB	76.9	76.0	73.3	69.6	65.2	61.4	60.7	61.2	62.9	67.1	69.2	72.4	(24)
	10%	DB	99.6	100.0	99.3	95.6	87.8	80.3	80.7	83.6	89.9	98.0	100.0	100.8	(25)
		MCWB	77.2	75.9	73.7	69.6	64.8	61.1	59.9	60.3	62.8	67.2	69.2	72.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.5	81.4	79.8	76.1	74.5	71.0	69.3	70.7	75.6	78.2	80.8	(27)
		MCDB	97.1	94.4	91.2	87.9	80.7	77.8	76.7	80.7	83.6	89.4	90.8	94.6	(28)
	2%	WB	81.0	81.2	80.2	78.0	73.5	71.3	67.8	67.1	68.7	73.3	76.6	79.3	(29)
		MCDB	93.6	92.6	89.6	86.5	81.4	76.2	76.6	79.7	82.6	88.8	89.2	91.4	(30)
	5%	WB	80.1	80.3	79.4	76.6	71.2	68.5	65.4	65.2	67.2	71.8	75.0	78.2	(31)
		MCDB	92.2	91.6	88.9	85.8	80.3	75.8	76.0	78.7	82.0	87.9	88.4	90.4	(32)
	10%	WB	79.4	79.5	78.6	75.2	69.2	66.0	63.1	63.4	65.9	70.3	73.5	77.1	(33)
		MCDB	91.3	90.8	88.4	85.9	79.2	74.6	75.2	77.7	81.6	87.6	89.2	90.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.4	19.7	23.1	24.0	23.1	25.7	28.4	28.9	28.1	25.2	22.3	(35)
		MCDBR	23.8	24.9	25.2	26.6	27.8	25.6	28.9	31.7	32.8	31.8	29.3	27.1	(36)
	5% WB	MCWBR	7.1	7.2	8.1	9.1	10.4	10.5	11.6	12.4	11.8	10.5	9.2	8.3	(37)
		MCWBR	18.5	18.3	18.2	19.3	19.6	17.2	21.9	27.6	28.0	26.3	23.1	19.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.402	0.393	0.380	0.353	0.333	0.322	0.316	0.315	0.335	0.356	0.361	0.378	(40)
		taud	2.495	2.535	2.558	2.587	2.574	2.592	2.585	2.544	2.489	2.460	2.485	2.513	(41)
	Ebn at Noon		299	298	294	290	284	282	287	298	304	306	310	306	(42)
		Edh at Noon	37	35	33	30	28	26	27	30	35	37	37	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2361	2254	2017	1765	1477	1337	1453	1793	2118	2394	2550	2539	(44)
	RadStd		144	155	147	92	111	95	92	50	80	58	66	109	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.83	N/A	N/A	N/A	+0.86	+1.01	N/A	N/A	N/A	+284

Nomenclature: See separate page